

Natural language statement

Domain

GroundTruth

Theorem 10.14. If p is a prime and $0 \leq x < 1$ then $(\phi(x^p))/(\phi(x)^p) = 1 + \sum_{i=1}^{\infty} a_i x^i$ where the a_i are integers. Notation: Here $\phi(x) = \prod_{n=1}^{\infty} (1 - x^n)$ is Euler's product, convergent for $0 \leq x < 1$, and $p(n)$ denotes the number of partitions of n.	Number theory	<pre>theorem niven_zuckerman_10_14_euler_product_prime_power (p : N) (hp : p.Prime) (phi : R -> R) (hphi : forall x : R, 0 <= x -> x < 1 -> Tendsto (fun m => prod n in Finset.Icc 1 m, (1 - x ^ n)) atTop (nhds (phi x))) : exists a : N -> Z, forall x : R, 0 <= x -> x < 1 -> phi (x ^ p) / phi x ^ p = 1 + p * sum' i : N, (a (i + 1) : R) * x ^ (i + 1) := by sorry</pre>
Theorem 10.15. For $0 \leq x < 1$ we have $x\phi(x)^4 = \sum_{m=1}^{\infty} b_m x^m$ where the b_m are integers and $b_m \equiv 0 \pmod 5$ if $m \equiv 0 \pmod 5$. Notation: Here $\phi(x) = \prod_{n=1}^{\infty} (1 - x^n)$ is Euler's product, convergent for $0 \leq x < 1$, and $p(n)$ denotes the number of partitions of n.	Number theory	<pre>theorem niven_zuckerman_10_15_mod_five_coefficients (phi : R -> R) (hphi : forall x : R, 0 <= x -> x < 1 -> Tendsto (fun m => prod n in Finset.Icc 1 m, (1 - x ^ n)) atTop (nhds (phi x))) : exists b : N -> Z, (forall m : N, m % 5 = 0 -> (5 : Z) ∣ b m) /\ forall x : R, 0 <= x -> x < 1 -> x * phi x ^ 4 = sum' m : N, (b m : R) * x ^ m := by sorry</pre>
Theorem 10.16. We have $p(5m + 4) \equiv 0 \pmod 5$. Notation: Here $\phi(x) = \prod_{n=1}^{\infty} (1 - x^n)$ is Euler's product, convergent for $0 \leq x < 1$, and $p(n)$ denotes the number of partitions of n.	Number theory	<pre>noncomputable def partitionCount (n : N) : N := Nat.card (Nat.Partition n) theorem niven_zuckerman_10_16_ramanujan_congruence (m : N) : partitionCount (5 * m + 4) % 5 = 0 := by sorry</pre>
Lemma 11.2. The function $\tau(n)$, representing the number of positive divisors of n, satisfies the inequality $\tau(n) \leq 2\sqrt{n}$ for $n \geq 1$. Notation: $\tau(n)$ is the number of positive divisors of n.	Number theory	<pre>theorem niven_zuckerman_11_2_divisor_bound (n : N) (hn : 1 <= n) : (n.divisors.card : R) <= 2 * Real.sqrt n := by sorry</pre>
Theorem 11.3. We have $\sum_{n=1}^{\infty} (\mu(n))/(n^2) \sum_{n=1}^{\infty} (1)/(n^2) = 1$. Notation: μ is the Mobius function.	Number theory	<pre>theorem niven_zuckerman_11_3_moebius_zeta_product : (sum' n : N, if n = 0 then 0 else (ArithmeticFunction.moebius n : R) / (n : R) ^ 2) * (sum' n : N, if n = 0 then 0 else 1 / (n : R) ^ 2) = 1 := by sorry</pre>
Corollary 11.4. We have $\sum_{n=1}^{\infty} (\mu(n))/(n^2) = (6)/(\pi^2)$. Notation: μ is the Mobius function; the proof uses the classical evaluation $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$.	Number theory	<pre>theorem niven_zuckerman_11_4_moebius_sum_eq_six_div_pi_sq : (sum' n : N, if n = 0 then 0 else (ArithmeticFunction.moebius n : R) / (n : R) ^ 2) = 6 / Real.pi ^ 2 := by sorry</pre>
An integer is square-free if it is divisible by no perfect square $a^2 > 1$. Theorem 11.5. The set of square-free integers has natural density $6/\pi^2$. Notation: Definition 11.1. If A is a set of positive integers and $A(n)$ denotes the number of elements of A not exceeding n, the "asymptotic" (or natural) density of A is $\delta(A) = \lim_{n \rightarrow \infty} A(n)/n$ when the limit exists. Definition 11.2. The "Schnirelmann density" $d(A)$ of a set A of non-negative integers is $d(A) = \inf_{n \geq 1} A(n)/n$. Definition 11.3. Assume 0 in A and 0 in B. The sum $A + B$ is the collection of all integers of the form $a + b$ where a in A and b in B.	Number theory	<pre>def HasNaturalDensity (A : Set N) (d : R) : Prop := Tendsto (fun n : N => (countingFunction A n : R) / n) atTop (nhds d) theorem niven_zuckerman_11_5_squarefree_density : HasNaturalDensity {n : N Squarefree n} (6 / Real.pi ^ 2) := by sorry</pre>
Lemma 11.6. Let $\sum c_j$ be a divergent series with $0 < c_j < 1$ for $j = 1, 2, \dots$. Then, given any real number $\epsilon > 0$, there is an integer N such that $\prod_{j=1}^n (1 - c_j) < \epsilon$ for every integer $n \geq N$. Notation: Divergence of $\sum c_j$ means that the partial sums tend to $+\infty$; since all factors $1 - c_j$ lie in $(0, 1)$ the partial products are decreasing, so the conclusion says exactly that they tend to 0.	Number theory	<pre>theorem niven_zuckerman_11_6_divergent_product_tendsto_zero (c : N -> R) (hpos : forall j, 0 < c j) (hlt : forall j, c j < 1) (hdiv : Tendsto (fun n => sum j in Finset.range n, c j) atTop atTop) : Tendsto (fun n => prod j in Finset.range n, (1 - c j)) atTop (nhds 0) := by sorry</pre>
Theorem 11.8. Let k be a fixed positive integer. If each integer in a set A is divisible by k or fewer distinct prime factors, then $\delta(A) = 0$. Notation: Definition 11.1. If A is a set of positive integers and $A(n)$ denotes the number of elements of A not exceeding n, the "asymptotic" (or natural) density of A is $\delta(A) = \lim_{n \rightarrow \infty} A(n)/n$ when the limit	Number theory	<pre>def HasNaturalDensity (A : Set N) (d : R) : Prop := Tendsto (fun n : N => (countingFunction A n : R) / n) atTop (nhds d) theorem niven_zuckerman_11_8_few_prime_factors_density_zero (k : N) (hk : 1 <= k) (A : Set N) (hA : forall n in A, n.primeFactors.card <= k) :</pre>

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exists. Definition 11.2. The *Schnirelmann density* $d(A)$ of a set A of non-negative integers is $d(A) = \inf_{\{n \geq 1\}} A(n)/n$. Definition 11.3. Assume 0 in A and 0 in B . The sum $A + B$ is the collection of all integers of the form $a + b$ where a in A and b in B .		HasNaturalDensity A 0 := by sorry
The alphabeta theorem of H. B. Mann. If A and B are sets of non-negative integers, each containing 0 , and if α , β , are the Schnirelmann densities of A , B , $A+B$, then $\geq \min(1, \alpha+\beta)$. In other words $\geq \alpha + \beta$ unless $\alpha + \beta \geq 1$, in which case $= 1$. Notation: Definition 11.1. If A is a set of positive integers and $A(n)$ denotes the number of elements of A not exceeding n , the *asymptotic* (or natural) density of A is $\delta(A) = \lim_{n \rightarrow \infty} A(n)/n$ when the limit exists. Definition 11.2. The *Schnirelmann density* $d(A)$ of a set A of non-negative integers is $d(A) = \inf_{\{n \geq 1\}} A(n)/n$. Definition 11.3. Assume 0 in A and 0 in B . The sum $A + B$ is the collection of all integers of the form $a + b$ where a in A and b in B .	Number theory	theorem niven_zuckerman_11_mann_alpha_beta_theorem (A B : Set N) [DecidablePred (. in A)] [DecidablePred (. in B)] [DecidablePred (. in {n : N exists a in A, exists b in B, n = a + b})] (hA : (0 : N) in A) (hB : (0 : N) in B) : min 1 (schnirelmannDensity A + schnirelmannDensity B) <= schnirelmannDensity {n : N exists a in A, exists b in B, n = a + b} := by sorry